Chapter 10 — Electrochemistry

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. Oxidation is:

(a) Gain of electrons **(b)** Loss of electrons **(c)** Loss of neutrons **(d)** Gain of protons

2. Reduction is:

(a) Gain of neutrons **(b)** Gain of electrons **(c)** Loss of electrons **(d)** Loss of protons

3. Oxidation occurs at:

(a) Salt bridge **(b)** Wire only **(c)** Cathode **(d)** Anode

4. Reduction occurs at:

(a) Cathode **(b)** Voltmeter **(c)** Anode **(d)** Salt bridge

5. Electrolytic cell converts:

(a) Heat to light **(b)** Chemical to electrical **(c)** Electrical to chemical **(d)** Light to heat

6. Galvanic cell converts:

(a) Light to chemical **(b)** Chemical to electrical **(c)** Electrical to chemical **(d)** Heat to electrical

7. In molten NaCl, cathode product:

(a) Cl_2 **(b)** Na **(c)** O_2 **(d)** H_2

8. In molten NaCl, anode product:

(a) Cl_2 **(b)** H_2 **(c)** Na **(d)** O_2

9. In aqueous NaCl, cathode product:

(a) Na **(b)** H_2 **(c)** Cl_2 **(d)** O_2

10. SHE potential is:

(a) +1.00 V **(b)** 0.00 V **(c)** -0.76 V **(d)** +0.34 V

11. SHE uses:

(a) Platinum black **(b)** Copper rod **(c)** Carbon only **(d)** Zinc rod

12. Cu_2+/Cu E°:

(a) -0.76 V **(b)** 0.00 V **(c)** +0.34 V **(d)** -2.38 V

13. Zn_2+/Zn E°:

(a) +0.80 V **(b)** +0.34 V **(c)** -0.76 V **(d)** 0.00 V

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14. In Cu-Zn cell, anode:

(a) H_2 **(b)** Zn **(c)** Cu **(d)** Salt bridge

15. Electrons flow from:

(a) Cathode to anode **(b)** Salt bridge to wire **(c)** Solution to wire **(d)** Anode to cathode

16. Salt bridge maintains:

(a) Pressure **(b)** Electrical neutrality **(c)** Temperature **(d)** Colour

17. Positive E° means greater tendency for:

(a) Hydrolysis **(b)** Reduction **(c)** Oxidation **(d)** Precipitation

18. Nernst equation relates E to:

(a) Colour **(b)** Ion concentration **(c)** Volume only **(d)** Mass only

19. More reactive metal can displace:

(a) Water only **(b)** Non-metal only **(c)** Less reactive metal **(d)** More reactive metal

20. Photovoltaic cell converts:

(a) Chemical to heat **(b)** Electricity to light **(c)** Heat to chemical **(d)** Light to electricity

21. DO means:

(a) Dry oxygen **(b)** Double oxide **(c)** Dissolved oxygen **(d)** Dissolved ozone

22. BOD means:

(a) Bond order data **(b)** Buffer oxygen dose **(c)** Biochemical oxygen demand **(d)** Basic oxide demand

23. High BOD indicates:

(a) High DO always **(b)** Organic pollution **(c)** No microbes **(d)** Pure water

24. Winkler method measures:

(a) Dissolved oxygen **(b)** pH only **(c)** Hardness only **(d)** Chloride only

25. Disproportionation means:

(a) No redox **(b)** Only oxidation **(c)** Only reduction **(d)** Same species oxidized and reduced

26. Oxidation number of free element:

(a) +1 **(b)** 0 **(c)** +2 **(d)** -1

27. Oxidation number of O usually:

(a) -2 **(b)** +2 **(c)** -1 **(d)** 0 always

28. Oxidation number of H usually:

(a) +1 **(b)** -1 **(c)** 0 **(d)** +2

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29. A stronger oxidizing agent has:

(a) Low solubility **(b)** More positive E° **(c)** More negative E° **(d)** Zero mass

30. Cell potential positive means reaction:

(a) Feasible **(b)** At equilibrium always **(c)** Non-feasible **(d)** No electrons

31. Oxidation is:

(a) Gain of electrons **(b)** Gain of protons **(c)** Loss of electrons **(d)** Loss of neutrons

32. Reduction is:

(a) Gain of neutrons **(b)** Loss of electrons **(c)** Loss of protons **(d)** Gain of electrons

33. Oxidizing agent is:

(a) Oxidized **(b)** Always gas **(c)** Always metal **(d)** Reduced

34. Reducing agent is:

(a) Reduced **(b)** Oxidized **(c)** Always non-metal **(d)** Always liquid

35. Oxidation number of free element is:

(a) +2 **(b)** -1 **(c)** 0 **(d)** +1

36. Disproportionation means same species is:

(a) Only reduced **(b)** Not changed **(c)** Oxidized and reduced **(d)** Only oxidized

37. Electrolytic cell converts:

(a) Light to heat **(b)** Heat to light **(c)** Chemical to electrical **(d)** Electrical to chemical

38. In electrolysis, oxidation occurs at:

(a) Wire **(b)** Anode **(c)** Salt bridge **(d)** Cathode

39. In electrolysis, reduction occurs at:

(a) Anode **(b)** Wire **(c)** Salt bridge **(d)** Cathode

40. Molten NaCl cathode product:

(a) Cl_2 **(b)** H_2 **(c)** O_2 **(d)** Na

41. Aqueous NaCl cathode product:

(a) O_2 **(b)** Na **(c)** Cl_2 **(d)** H_2

42. Charge passed equals:

(a) $I^+ t$ **(b)** I/t **(c)** It **(d)** t/I

43. One faraday is about:

(a) 6.02×10^{23} C **(b)** 1.6×10^{-19} C **(c)** 22.4 C **(d)** 96500 C mol^{-1}

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44. Electrical double layer forms at:

(a) Wire only **(b)** Solid lattice only **(c)** Metal-solution interface **(d)** Gas phase only

45. SHE potential is:

(a) -0.76 V **(b)** +1.00 V **(c)** +0.34 V **(d)** 0.00 V

46. More positive E° means greater tendency for:

(a) Oxidation **(b)** Evaporation **(c)** Precipitation **(d)** Reduction

47. Galvanic cell converts:

(a) Heat to light **(b)** Light to heat **(c)** Electrical to chemical **(d)** Chemical to electrical

48. Salt bridge maintains:

(a) Electrical neutrality **(b)** Temperature **(c)** Colour **(d)** Pressure

49. Nernst equation relates E to:

(a) Density **(b)** Ion concentration **(c)** Colour **(d)** Molar mass only

50. Photovoltaic cell converts:

(a) Chemical to electrical **(b)** Heat to light **(c)** Light to electrical **(d)** Electrical to chemical

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Answer Key with Brief Reasons

1. Answer: (b) Loss of electrons — Redox definition.

2. Answer: (b) Gain of electrons — Redox definition.

3. Answer: (d) Anode — Anode oxidation.

4. Answer: (a) Cathode — Cathode reduction.

5. Answer: (c) Electrical to chemical — Non-spontaneous driven.

6. Answer: (b) Chemical to electrical — Spontaneous redox.

7. Answer: (b) Na — Na^+ reduced.

8. Answer: (a) Cl_2 — Cl^- oxidized.

9. Answer: (b) H_2 — Water reduced more easily.

10. Answer: (b) 0.00 V — Reference electrode.

11. Answer: (a) Platinum black — Inert catalytic surface.

12. Answer: (c) +0.34 V — Standard value.

13. Answer: (c) -0.76 V — Standard value.

14. Answer: (b) Zn — Zn oxidized.

15. Answer: (d) Anode to cathode — External circuit.

16. Answer: (b) Electrical neutrality — Ion flow.

17. Answer: (b) Reduction — Relative to SHE.

18. Answer: (b) Ion concentration — Concentration effect.

19. Answer: (c) Less reactive metal — Activity series.

20. Answer: (d) Light to electricity — Solar cell.

21. Answer: (c) Dissolved oxygen — Water quality.

22. Answer: (c) Biochemical oxygen demand — Pollution measure.

23. Answer: (b) Organic pollution — Microbes consume O_2.

24. Answer: (a) Dissolved oxygen — Water analysis.

25. Answer: (d) Same species oxidized and reduced — Cl_2 in alkali.

26. Answer: (b) 0 — Rule.

27. Answer: (a) -2 — Rule except peroxides.

28. Answer: (a) +1 — Rule except metal hydrides.

29. Answer: (b) More positive E° — Accepts electrons.

30. Answer: (a) Feasible — Spontaneous redox.

31. Answer: (c) Loss of electrons — OIL RIG.

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32. Answer: (d) Gain of electrons — OIL RIG.

33. Answer: (d) Reduced — Accepts electrons.

34. Answer: (b) Oxidized — Donates electrons.

35. Answer: (c) 0 — Elemental state.

36. Answer: (c) Oxidized and reduced — Dual role.

37. Answer: (d) Electrical to chemical — Non-spontaneous driven.

38. Answer: (b) Anode — Anode definition.

39. Answer: (d) Cathode — Cathode definition.

40. Answer: (d) Na — Na^+ reduced.

41. Answer: (d) H_2 — Water reduced.

42. Answer: (c) It — Q=It.

43. Answer: (d) 96500 C mol^{-1} — Mole of electrons.

44. Answer: (c) Metal-solution interface — Charge separation.

45. Answer: (d) 0.00 V — Reference electrode.

46. Answer: (d) Reduction — Gain electrons.

47. Answer: (d) Chemical to electrical — Spontaneous redox.

48. Answer: (a) Electrical neutrality — Ion flow.

49. Answer: (b) Ion concentration — Concentration effect.

50. Answer: (c) Light to electrical — Solar cell.

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